Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/20/2005
PI Name:	Niesel, David William Ph.D.		
Project Title:	Streptococcus pneumoniae Gene Expression and Virulence Potential in the Space Environment		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77555-1019	**Congressional District:**	14
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-02
Start Date:	02/01/2000	**End Date:**	11/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	renewed. Updated Aug. 26, 2005.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Watson, David (The University of Texas Medical Branch)		
Grant/Contract No.:	NCC2-1160		
Performance Goal No.:			
Performance Goal Text:			
	Extended spaceflight and the long-term habitation of the space environment pose a challenge in the protection of humans from infectious disease. While evidence is accumulating about space induced alterations in the immune system which may render personnel more susceptible to microbial infections, relatively little is known as to the effect of the space environment on the physiology, gene expression or the virulence potential of opportunistic pathogenic bacteria. S. pneumoniae is both a commensal inhabitant of the nasopharyx and is an opportunistic bacterial pathogen in individuals whose immune systems are impaired. A mediator of serious disease (pneumonia, otitis media, and meningitis), S. pneumoniae is easily spread by aerosols. Carriage rates in healthy individuals vary from 5-70% and this microbe has been isolated from crewmembers immediately prior to flight. S. pneumoniae and other opportunistic bacterial pathogens could represent a significant threat to personnel during long term habitation of space or in extended spaceflight. It is important to understand the effects of microgravity and other elements of the space environment on the virulence		

Task Description:	potential of opportunistic microbes and understand their interaction with the host to generate a rationale risk assessment for infectious disease threats to space personnel. In this study, we are investigating Streptococcus pneumoniae gene and protein expression and assessing the virulence potential of this opportunistic pathogen under conditions of low shear modeled microgravity (LSMMG) and in the space environment. The experimentation is organized into three specific objectives. Experiments are in progress to identify and characterize S. pneumoniae genes and proteins which are differentially expressed in response to the space environment or in response to LSMMG using high aspect rotating vessels. The second objective is to investigate the expression of known S. pneumoniae virulence genes and properties in LSMMG. Finally, we will determine if S. pneumoniae genes differentially expressed in LSMMG or during space flight are also expressed during animal infection. This experimental approach will result in new information about a significant human pathogen, add to our knowledge about the S. pneumoniae pathogenic mechanism and also provide basic information on a bacterial model system of gene and protein expression using a microgravity analogue and in the space environment. These studies will also provide important information that can be used to develop a rationale risk assessment for this organism and other opportunistic microbes in the context of long-term space flight and habitation. In addition, these studies will contribute to our understanding of the virulence of this bacterium on earth. Thus, this experimentation will benefit NASA objectives to advance manned space exploration, public health agencies and public and private commercial entities.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Experimentation conducted for this project will lead to an increased understanding of the pathogenic mechanism of S. pneumoniae. This bacterium is an opportunistic human pathogen which is the number one mediator of community acquired pneumonia and the second leading cause of meningitis world-wide. It contributes to respiratory infections which are the leading cause of childhood death in the world today. Our results will advance our understanding of the pathogenic mechanism of this important pathogen. New knowledge about the pathogenic mechanism will have direct earth benefits in terms of understanding the disease mechanism(s) of S. pneumoniae and could lead to new diagnostics, vaccines and therapeutics.
Task Progress:	This past year, we continued in our efforts to develop alternative stowage and storage parameters for our flight experiment to attempt to take advantage alterative launch opportunities. A major component of our activity was directed at understanding the limitations of using lyophilized bacterial cultures in an automated hardware configuration that permitted sampling of the bacterial culture at different times and phases of growth. This analysis identified multiple challenges associated with the use of lyophilized bacterial cultures in this hardware. Issues related to viability and storage requirements for S. pneumoniae before activation (up to 3 months), premature distribution of the lyophilized bacterial cells in the closed loop system, growth of environmental bacteria in the humid atmosphere of the sealed cassettes and water vapor permeability of cassette tubing and culture bags emerged as key concerns of these studies. For the rest of the year, efforts focused on developing experimental protocols for both long and short-term missions for either an ISS mission or Shuttle sortie, respectively. We continued training with equipment to prepare samples for launch and post-flight analysis. The experiment is pending manifesting on a shuttle mission early in 2006. New hardware for the experiment is under consideration and optimization and protocols for enhanced science recovery are also in progress. We are using S. pneumoniae strain Spn4 for all flight and low shear modeled microgravity (LSMMG) experiments. The complete annotated genomic sequence of S. pneumoniae Spn4 is available in databases at Genbank and at The Institute for Genomic Research. This allows direct sequence comparison to identify genes expressed in LSMMG or in spaceflight conditions. We have obtained S. neumoniae total genomic microarrays from TIGR and have optimized protocols to investigate S. pneumoniae gene expression during this past year. We have completed a pilot study to assess S. pneumoniae gene expression at different temperatures to become familiar with the DNA microarray chip protocols and data analysis as well as clustering components of the Spotfire computational software. In ground-based studies, gene expression was compared between S. pneumoniae cultured under LSMMG and laboratory controls to identify differentially regulated genes. For LSMMG conditions, S. pneumoniae was grown to mid-logarthmic phase in rotating wall bioreactors (HARVs). RNA was isolated and analyzed using genomic microarrays with results verified by quantitative RT-PCR. Overall, 326 genes were differentially expressed under LSMMG versus normal gravity (1xg) of which 33 genes were up-regulated and 293 were down-regulated 1.5-fold or higher. Differentially expressed genes were from various functional groups including amino acid synthesis, cellular metabolism, cell structure, protein production/degradation, transport and binding, hypothetical proteins, and proteins of unknown function. The majority of differentially regulated genes represented hypothetical proteins followed by genes encoding transport and binding proteins and various metabolic and cellular components. Down-regulated genes involved those functioning in competence, antimicrobial peptide production, and transport and binding. Genes showing increased expression included those functioning in amino acid synthesis, metabolism, protein synthesis, and transport and binding. A known virulence factor was up-regulated 3.4-fold under LSMMG culture conditions. Several differentially regulated genes were identical to Salmonella genes shown to be altered under LSMMG culture conditions. The response of S. pneumoniae to growth in LSMMG demonstrates the extensive range of physiological alterations that take place as this bacterium adapts to this unique environment. Such genetic changes can reveal differential expression patterns and potential regulatory pathways to better understand the physiological response of S. pneumoniae and other bacteria in response to microgravity. The ultimate goal of these experiments will be the comparison of these ground-based microgravity analogue results with those from cultures after space flight.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
Abstracts for Journals and Proceedings	Niesel, W. D. "Streptococcus pneumoniae alters Protein Profiles and Virulence Potential in Response to Low Shear Modeled Microgravity" ASM Annual Meeting, New Orleans, LA ASM , May-2004
Abstracts for Journals and Proceedings	Allen, C.A. U. Pandya, D.A. Watson, and D.W. Niesel "A Global Transcriptional Profile of Streptococcus pneumoniae in Response to Modeled Microgravity. " ASM General Meeting, New Orleans, LA ASM , May-2004

Abstracts for Journals and Proceedings	Pandya, U., Allen, C., Watson and Niesel D. W. "Global profiling of Streptococcus pneumoniae transcription as a function of growth temperature." ASM General Meeting, New Orleans, LA ASM , May-2004
Abstracts for Journals and Proceedings	Pandya, U., Dang, Y. D., Wilson, C.J., Watson, D. A and Niesel, D.W. "Identification of a Histidine Kinase in Streptococcus pneumoniae that is differentially expressed at 33ºC and 37ºC" ASM General Meeting, Washington, D.C ASM , May-2003
Abstracts for Journals and Proceedings	Pandya,U. D.W. Niesel and Dang, T. "SELDI Protein Expression Profiling and Biomarker Identification of Streptococcus pneumoniae. " ASMS Annual Meeting, Orlando, FL ASMS , Apr-2003
Awards	Orihuela, C. "Texas Space Grant Consortium Pre-doctoral Scholarship" Jan-2003
Awards	Allen, C.A. "Texas Space Grant Consortium Pre-doctoral Scholarship" Jul-2004
Awards	Allen, C.A. "NASA - Graduate Student Research Program" Pre-doctoral Fellowship" Aug-2004
Presentation	Niesel, D. W. , "Streptococcus pneumoniae Response to Low Shear Modeled Microgravity Alters Interactions with the Host" 15th Humans in Space Symposium, Graz, Austria May-2005
Presentation	Niesel, D. W. "S. pneumoniae alters Protein Profiles and Virulence Potential in Response to Modeled Microgravity. " ASM General Meeting, New Orleans, LA May-2003
Presentation	Niesel, D. W. "Streptococcus pneumoniae Gene Expression and Virulence Potential in Low Shear and Modeled Microgravity Environments (LSMMG) " NAS Workshop, Woods Hole, MA, Apr-2004
Presentation	Niesel, D. "Streptococcus pneumoniae in the Space Environments. " Depatmetn of Biology, St. Mary's University, San Antonio, TX Apr-2003